



Bambu AMS No-Catch Y-Splitter PC4-M10



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[VIEW IN BROWSER](#)

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Summary

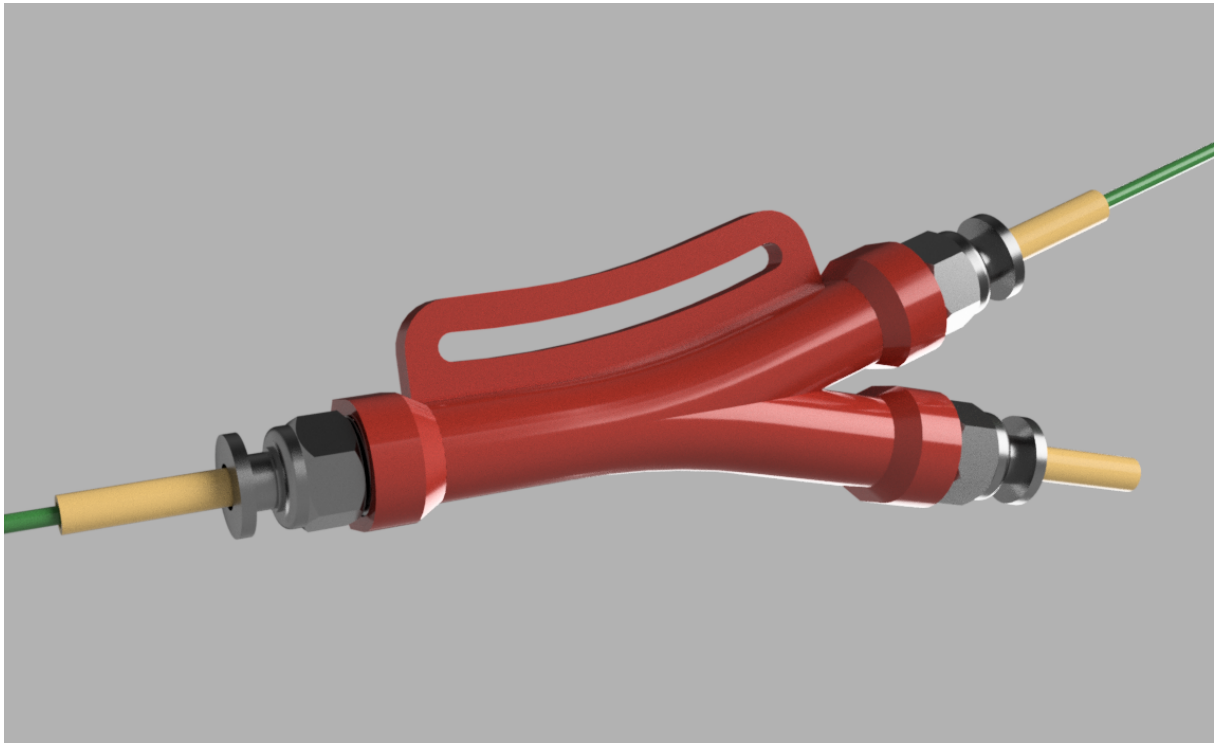
This is a new and improved "no catch" y-splitter for Bambu Lab printers with AMS units.

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Tags: [cap](#) [cover](#) [ptfe](#) [bambulab](#) [bambulabx1carbon](#) [ams](#) [x1c](#) [x1carbon](#) [x1carbonams](#) [p1p](#) [p1s](#)

Update 2023-11-07: Added a "long slot" version to accommodate some inconsistencies in the length of the PTFE tube sticking out the back of the printer that feeds the hot end.

- I want to thank users [@xenon2008](#) and [@Tvorim3D](#) for leaving a review noting that the PTFE tube going to the hot end was longer on their printers and didn't allow the slot in the splitter to align with the screw holes on the back of the printer. Hopefully this update will address the issue. I only modified the slot, making it extend along the full length of the splitter. There is no change to the internals of the splitter itself. 29 new files (mostly various thread clearances and mounting screw sizes), including a self-tap version STEP file and my Fusion 360 source file have been added.

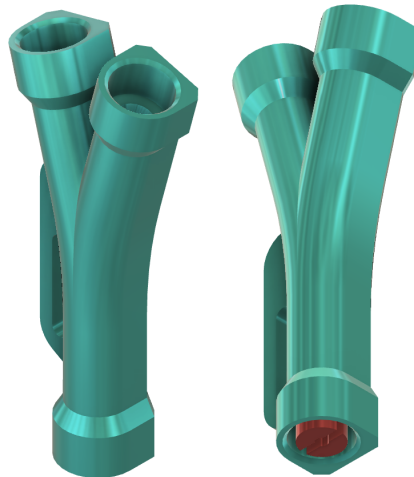
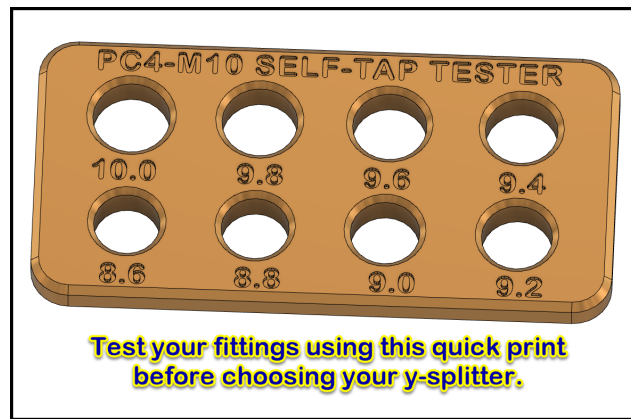


Update 2023-09-01: Fixed the reversed threads on one of the sockets for the PC4-M10 fittings.

- Big thanks to [@skeybord](#) for catching this error. All files have been replaced with corrected versions, including the Fusion 360 source file.

Update 2023-08-20: Added a set of self-tapping splitters and a tester to determine the best size opening for your fittings.

- Based on user feedback and my own experience, it seems that not all PC4-M10 fittings are made with the same thread specs. Some fittings seem to have finer thread pitch than others and the length of the threaded portion can vary as well.
- I have added a set of splitters that do not have the threads modeled into them so they can self-tap as you screw them in. I included a test print so you can figure out the correct size opening to choose before printing your splitter. A spec I found for tap size for M10 holes is 8.5 mm. The tester has hole sizes ranging from 8.6mm to 10mm in 0.2mm increments. I recommend printing with PETG instead of PLA since it is a bit more flexible and you should have any easier time screwing in your fitting.
- A huge thanks goes out to [@marbledradial](#) for reaching out and working with me to test the self-tapping splitters!



Update 2023-08-17: Added complete details for assembly and installation.

(sorry for the initial oversight!)

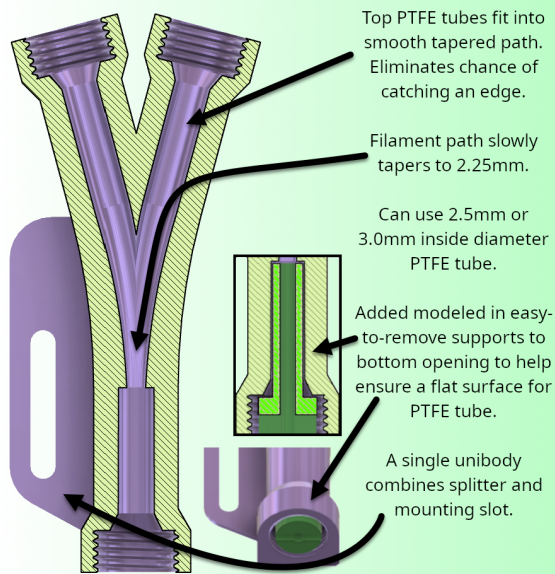
This is an upgrade from my original y-splitter ([Y-Splitter for the Bambu Lab X1C AMS](#)) .

Although my original design works very well, this improved version requires little to no post-processing, is easier to print, and the filament has a minimal chance of catching on anything as it passes through the splitter on its way to the hot end.

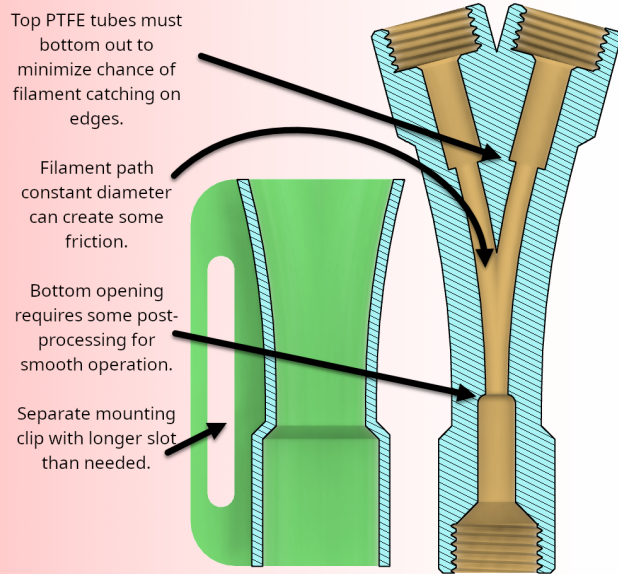
This design is also built with user parameters to make small tweaks easier. I have included my Fusion 360 source files so you can edit and remix for yourself.

Once I realized that the filament really only needs to slide easily along the path **into** the hot end, I was able to remove most of the pain points that required some fiddling after printing. After a print is completed, there is zero chance the filament will catch on anything as it is backed away from the hot end and out through the y-splitter. Here is a cross-section view comparing the old and new versions.

New Version

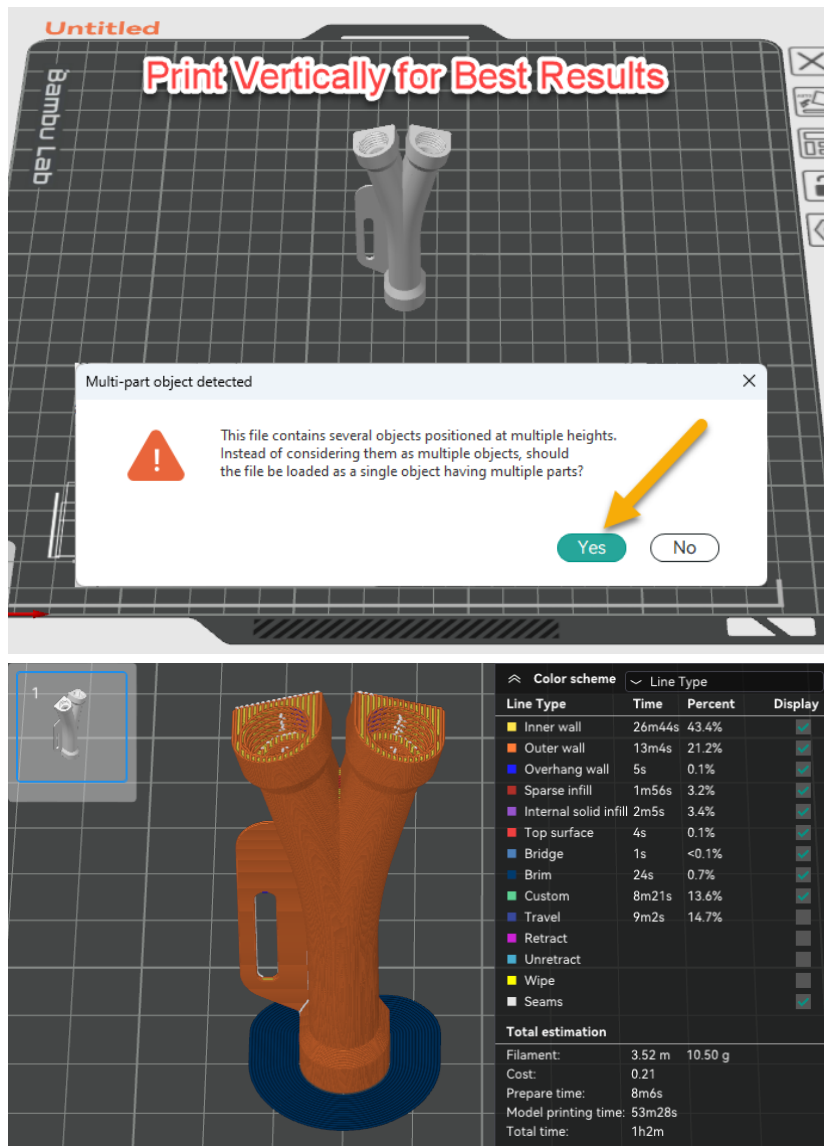


Old Version



Printing Tips

I got the best results with my X1C using a 0.20 profile and PETG filament. I highly recommend printing with the Y standing up vertically with the single leg on the build plate. I tried printing in laying down flat, but did not get good results due to the PTFE and filament paths not coming out perfectly round.



I use a slightly modified stock profile. I printed with the Arachne wall generator, an 8mm outer brim, 4 walls, 4 top and bottom layers, and slowed down outer walls to 100mm/s. I also tried different seam positions and settled on "Back".

Tools and Parts You Will Need

Tools:

- 2mm hex wrench
- 2.5mm hex wrench
- PTFE tube cutter or utility knife with new blade
- 10mm wrench or adjustable wrench
- Calipers or ruler
- Small chamfering drill bit or diamond coated needle file

Parts:

- 3D printed y-splitter
- M3 x 6mm or M3 x 8mm socket head screw (to match the version you print)
- 3 x PC4-M10 PTFE connectors
- 1 length of PTFE tube cut to 45mm - 50mm
- 1 length of PTFE tube cut to approximately 100mm
 - Note 1: PTFE tube must be 4mm outside diameter x 2.5mm inside diameter. This is the same size used and provided as a spare by Bambu Lab.
 - Note 2: PTFE tubes with a 3mm inside diameter should also work. See the details in the Post-Processing section below.
- 1 length of filament ~150mm long for testing

Assembly and Installation

Step 1 - Disconnect the PTFE Tube

Disconnect the PTFE tube from the rightmost side of the coupler by depressing the spring loaded fitting and pulling the tube out. You will be reconnecting this to one side of the y-splitter later.

Step 2 - Cutting PTFE Tubes

It's best to use a PTFE tube cutter if you have one, but if not, it's fine to use a utility knife. Make sure you have a sharp blade and make sure to make the cut as square as possible.

Cut a length of PTFE tube 45mm-50mm in length from the spare PTFE tube that came with your printer. It is important to use a tube that is 4mm OD x 2.5mm ID. Most common PTFE tubes for 1.75mm filament have an inside diameter of 2.0mm. We need the larger inside diameter to ensure the filament moves freely through the tube. This piece will be fitted to the single leg on the y-splitter (the output side).

Cut another length of PTFE tube 100mm for the other side of the "Y" side of the splitter. The length here is less critical and depending on your setup, you may want to use more or less.

Step 3 - Install the Threaded PC4-M10 Connectors

Being careful not to cross-thread, screw in the 3 PC4-M10 connectors into the splitter. Hand tighten them as far as you can and then fully secure them with the M10 or adjustable wrench. Don't over tighten or you may strip the threads.

Step 4 - Insert PTFE Tubes

Insert the short piece of PTFE tube you cut firmly into the single leg side of

the y-splitter. Make sure the tube bottoms out in the hole. This will help insure the filament won't catch on the edge of the tube as it passes through.

Insert the second piece of PTFE tube you cut into the upper fitting. This path is tapered, so there you just need to insert it until you feel some resistance. It shouldn't happen, but make sure you don't end up inserting it past the intersection of the 2 legs in the middle of the splitter.

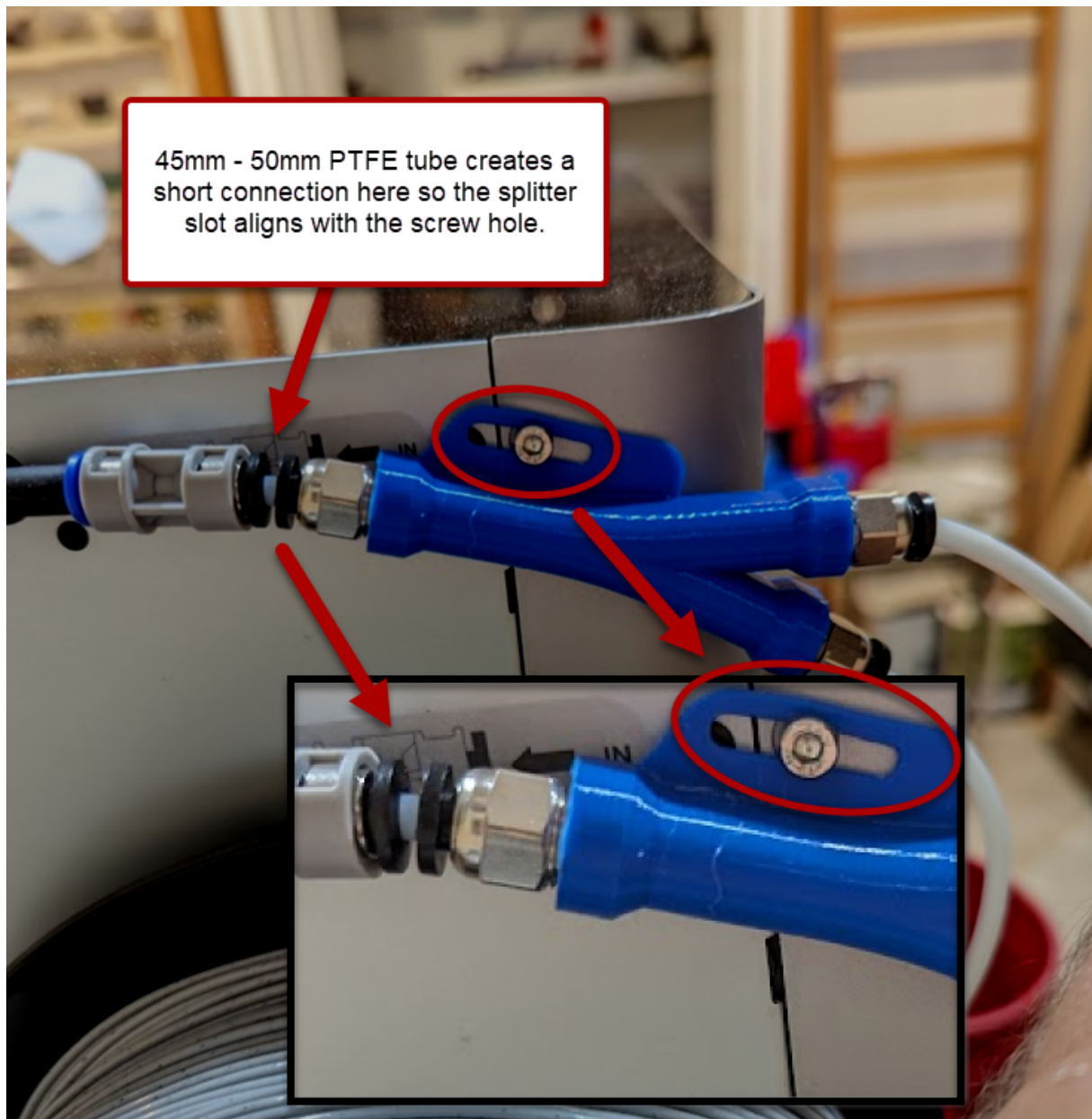
This is a good time to test the operation of the splitter. Take a short piece of filament with the end cut square and slide it into the tube on the right side. Push it until it slides out the left side. It should move through the splitter easily with little resistance. Repeat the same test pushing the filament through the leg without the PTFE tube. It should also move through the splitter without resistance.

If you do feel the filament "catch an edge", see the **Post-Processing** section below for some suggestions.

Once everything is moving freely, you can insert the PTFE tube you disconnected in Step 1 into the bottom leg of the "Y", again making sure it is inserted until you feel resistance, but not past the intersection of the 2 legs in the middle of the splitter. You will want to make sure this tube is routed under the spool holder.

Step 5 - Mounting the Splitter

Insert the PTFE tube on the left side into the coupler feeding the extruder. This connection has a very short amount of travel before it bottoms out. You should only have a few millimeters of PTFE tube visible between the coupler and the y-splitter.



Using a 2mm hex wrench, remove the right-most screw from the corner radius on the cover. Using a 2.5mm hex wrench, insert a M3 x 6mm (or M3 x 8mm if using the thicker version) socket head screw through the slot and secure the clip to the corner hole. The screw bottoms out quickly in that hole. That is why there are versions for 6mm and 8mm mounting screws.

Step 6 - Test the AMS

You can test that the filament loads and unloads easily from the AMS by doing a couple of Load/Unload operations from the printer's touchscreen.

Step 7 - Test the Spool Holder Filament

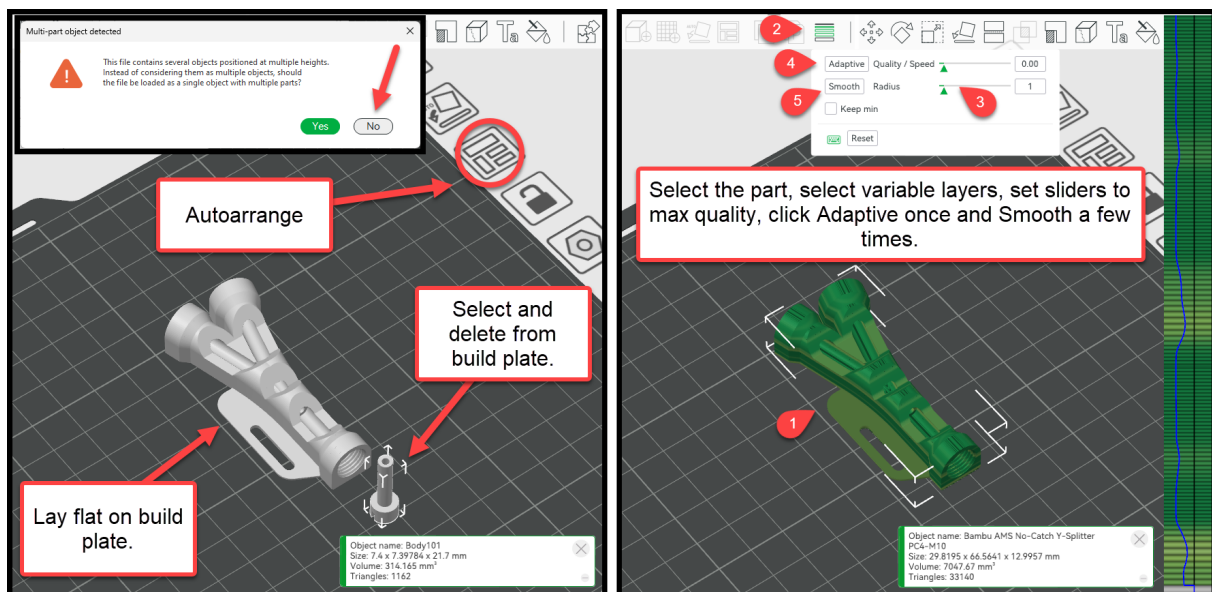
To test the filament loading and unloading from the spool holder, make sure the filament is unloaded from the AMS (there is no filament visible in the PTFE tube coming from the AMS).

Make sure the end of the filament on the spool holder is cut square and is feeding from the bottom of the spool (it will turn counter-clockwise as the filament is fed into the extruder). Insert the filament into the short PTFE tube on the splitter and push it all the way in until it hits the extruder. Using the touchscreen menu, perform a load operation by selecting the spool holder location on the AMS tab. The printer will display prompts to guide you.

Update 2023-08-04: Print Without Modeled-In Support

Based on a tip from user [@roadshow](#) in the comments below, you can get good results printing the splitter laying flat and using variable layer height. Setting both sliders to the highest quality and adding several "smoothing" clicks gave me a very good result as well using PETG ([@roadshow](#) used ASA). Thanks to [@roadshow](#) for testing and figuring out settings for the best results!

If you do print it flat, say "No" to the prompt to load as a single object. Then use auto arrange to see the separate support body and delete it from the build plate. Lay the splitter on the flat face and slice. I used the same profile as before, but added the variable layer height steps.



Post-Processing

The only bit of post-processing you may need is to gently clean out the 2.25mm opening at the bottom of the filament path. It is important to not be overly aggressive here because you want to keep the opening smaller than the 2.5mm opening in the PTFE tube. I have a small hand drill set that has a 2.2mm drill bit. I used that to gently clean out the hole at the bottom of the filament path. A small diamond coated needle file with a light touch also works. Whatever you use, be careful not to open the hole too much or you will create an opportunity for the filament to catch on the

PTFE tube if you end up opening the hole larger than 2.5mm. The goal is to clean up the top edges of the hole to help the filament move smoothly through the opening.

The support that is modeled into the bottom should come out cleanly by twisting it with a flat screwdriver to break it loose. I did model a hole through the center of the support in case it breaks off short so you can use a drill bit or something similar to remove any small segments left behind. Once removed, this exposes the face for the PTFE tube to fit against. You want the fit as tight and as flat as possible against the opening.

Gently clearing the opening and getting the support removed cleanly will help insure that the filament will pass through smoothly without catching any edges on the PTFE tube. Other than that, I needed no other post-processing.

This version should work with 2.5mm or 3.0mm inside diameter PTFE tubes. Printables user [Wgragg](#) was interested in using some PTFE tube with a 3.0mm inside diameter. I sent him an earlier iteration of this design for testing and he reported that it worked fine. I have not tested this personally, so YMMV.

If you would like the cap shown in the pictures for the PTFE tube when the spool holder is not being used, you can grab it here: [PTFE Cap for Bambu Lab AMS Y-Splitter](#)

Thanks!

I hope you find this useful. Please share your make and let me know in the comments if you have any questions.

Links to some items similar to what I used (not affiliate links):

Diamond Coated Needle File Set

<https://www.amazon.com/Yakamoz-Diamond-Triangular-Riffler-Jewelers/dp/B092D6KFN9>

Bambu Lab 4mm OD x 2.5mm ID PTFE tube

<https://us.store.bambulab.com/products/bambu-lab-ptfe-tube-3-pcs-370mm-1-550mm-2>

<https://us.store.bambulab.com/products/bambu-ptfe-tube-8pcs>

PC4-M10 PTFE Connector

<https://www.amazon.com/BIQU-Straight-Pneumatic-Connector-Extruder/dp/B01IB81IHG>

PTFE tube cutter

<https://www.amazon.com/PTFE-Teflon-cutter-tube-Allen/dp/B0756D2T73/>

Countersink drill bit set:

<https://www.amazon.com/dp/B07MCS2441?psc=1>

ADDED BONUS

Are you still reading this? ;-)

Update 2023-08-11: New Peek-A-Boo Versions Added

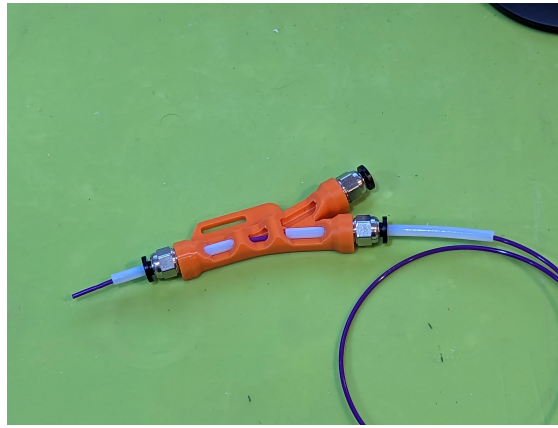
I just added a few more versions of the peek-a-boo splitter. There are now “tighter” thread versions with supports. I also added a **“Peek-A-Boo Splitters - no support - print flat”** folder with normal and tighter threads that have the support removed for printing flat.

Although I've been using the peek-a-boo splitter myself, I'm surprised that this little novelty splitter is so popular. Thanks to everyone for the positive feedback and support!

As an added bonus, I also included some “peek-a-boo” y-splitter models that started life as test prints so I could view and demonstrate a cross section view of the design. After playing around, I came up with a way to have partial cross-section segments that allow you to “peek” inside the splitter. Here is a small sample of my many test prints.



I have not fully tested these, but I can't see any reason they wouldn't work. I can't decide if they are cool or just goofy looking. I'll let you be the judge!



Model files



Self-Tap Splitters (added 2023-08-20)

19 files



pc4-m10-self-tap-tester.3mf

☐ Print this first to see which diameter works best with your fittings

no-catch-y-splitter-self-tap-10mm-m3x8-mount.step



no-catch-y-splitter-self-tap-100mm-m3x8-mount.3mf



no-catch-y-splitter-self-tap-98mm-m3x8-mount.3mf



no-catch-y-splitter-self-tap-96mm-m3x8-mount.3mf



no-catch-y-splitter-self-tap-94mm-m3x8-mount.3mf



no-catch-y-splitter-self-tap-92mm-m3x8-mount.3mf



no-catch-y-splitter-self-tap-90mm-m3x8-mount.3mf



no-catch-y-splitter-self-tap-88mm-m3x8-mount.3mf



no-catch-y-splitter-self-tap-86mm-m3x8-mount.3mf

no-catch-y-splitter-self-tap-10mm-m3x6-mount.step



no-catch-y-splitter-self-tap-100mm-m3x6-mount.3mf



no-catch-y-splitter-self-tap-98mm-m3x6-mount.3mf



no-catch-y-splitter-self-tap-96mm-m3x6-mount.3mf



no-catch-y-splitter-self-tap-94mm-m3x6-mount.3mf



no-catch-y-splitter-self-tap-92mm-m3x6-mount.3mf



no-catch-y-splitter-self-tap-90mm-m3x6-mount.3mf



no-catch-y-splitter-self-tap-88mm-m3x6-mount.3mf



no-catch-y-splitter-self-tap-86mm-m3x6-mount.3mf



Splitters for M3x6mm mounting screw

3 files



no-catch-y-splitter-pc4-m10-m3x6-mount-normal-threa... .3mf

☐ Updated 2023-09-01 - thread fix



no-catch-y-splitter-pc4-m10-m3x6-mount-tighter-thre... .3mf

☐ Updated 2023-09-01 - thread fix



no-catch-y-splitter-pc4-m10-m3x6-mount-looser-threa... .3mf

☐ Updated 2023-09-01 - thread fix



Splitters for M3x8mm mounting screw

3 files



no-catch-y-splitter-pc4-m10-m3x8-mount-looser-threa... .3mf

☐ Updated 2023-09-01 - thread fix



no-catch-y-splitter-pc4-m10-m3x8-mount-normal-threa... .3mf

📄 Updated 2023-09-01 - thread fix



no-catch-y-splitter-pc4-m10-m3x8-mount-tighter-thre... .3mf

📄 Updated 2023-09-01 - thread fix



Fusion 360 Source files

2 files

bambu-ams-no-catch-y-splitter-pc4-m10-20230901.f3d

📄 Updated 2023-09-01 - thread fix

no-catch-y-splitter-long-slot-pc4-m10-231107.f3d



Peek-A-Boo Splitters

6 files



no-catch-y-splitter-peek-a-boo-m3x8-mount-tighter-t... .3mf

📄 Updated 2023-09-01 - thread fix



no-catch-y-splitter-peek-a-boo-m3x6-mount-looser-th... .3mf

📄 Updated 2023-09-01 - thread fix



no-catch-y-splitter-peek-a-boo-m3x6-mount-tighter-t... .3mf

📄 Updated 2023-09-01 - thread fix



no-catch-y-splitter-peek-a-boo-m3x6-mount-normal-th... .3mf

📄 Updated 2023-09-01 - thread fix



no-catch-y-splitter-peek-a-boo-m3x8-mount-normal-th... .3mf

📄 Updated 2023-09-01 - thread fix



no-catch-y-splitter-peek-a-boo-m3x8-mount-looser-th... .3mf

📄 Updated 2023-09-01 - thread fix



Peek-A-Boo Splitters - no support - print flat

6 files



no-catch-y-splitter-peek-a-boo-m3x8-tighter-print-f... .3mf

📄 Updated 2023-09-01 - thread fix



no-catch-y-splitter-peek-a-boo-m3x6-looser-print-fl... .3mf

📄 Updated 2023-09-01 - thread fix



no-catch-y-splitter-peek-a-boo-m3x8-normal-print-fl... .3mf

📄 Updated 2023-09-01 - thread fix



no-catch-y-splitter-peek-a-boo-m3x6-tighter-print-f... .3mf

📄 Updated 2023-09-01 - thread fix



no-catch-y-splitter-peek-a-boo-m3x6-normal-print-fl... .3mf

📄 Updated 2023-09-01 - thread fix



no-catch-y-splitter-peek-a-boo-m3x8-looser-print-fl... .3mf

📄 Updated 2023-09-01 - thread fix



LONG SLOT Splitters for M3x8 mounting screw (2023-11-07) 3 files



no-catch-y-splitter-long-slot-m3x8-mount-normal-thr... .3mf



no-catch-y-splitter-long-slot-m3x8-mount-looser-thr... .3mf



no-catch-y-splitter-long-slot-m3x8-mount-tighter-th... .3mf



LONG SLOT Splitters for M3x6 mounting screw (2023-11-07) 3 files



no-catch-y-splitter-long-slot-m3x6-mount-normal-thr... .3mf



no-catch-y-splitter-long-slot-m3x6-mount-looser-thr... .3mf



no-catch-y-splitter-long-slot-m3x6-mount-tighter-th... .3mf



LONG SLOT Self-Tap Splitters (added 2023-11-07)

17 files

no-catch-y-splitter-self-tap-long-slot-10mm-m3x6-mo... .step



no-catch-y-splitter-self-tap-long-slot-100mm-m3x6-m... .3mf



no-catch-y-splitter-self-tap-long-slot-100mm-m3x8-m... .3mf



no-catch-y-splitter-self-tap-long-slot-98mm-m3x6-mo... .3mf



no-catch-y-splitter-self-tap-long-slot-98mm-m3x8-mo... .3mf



no-catch-y-splitter-self-tap-long-slot-96mm-m3x6-mo... .3mf



no-catch-y-splitter-self-tap-long-slot-96mm-m3x8-mo... .3mf



no-catch-y-splitter-self-tap-long-slot-94mm-m3x6-mo... .3mf



no-catch-y-splitter-self-tap-long-slot-94mm-m3x8-mo... .3mf



no-catch-y-splitter-self-tap-long-slot-92mm-m3x6-mo... .3mf



no-catch-y-splitter-self-tap-long-slot-92mm-m3x8-mo... .3mf



no-catch-y-splitter-self-tap-long-slot-90mm-m3x6-mo... .3mf



no-catch-y-splitter-self-tap-long-slot-90mm-m3x8-mo... .3mf



no-catch-y-splitter-self-tap-long-slot-88mm-m3x6-mo... .3mf



no-catch-y-splitter-self-tap-long-slot-88mm-m3x8-mo... .3mf



no-catch-y-splitter-self-tap-long-slot-86mm-m3x6-mo... .3mf



no-catch-y-splitter-self-tap-long-slot-86mm-m3x8-mo... .3mf

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- ✗ | Free Cultural Works
- ✗ | Meets Open Definition